USER MANUAL:

AUTOMATED SHOE POLISHING MACHINE

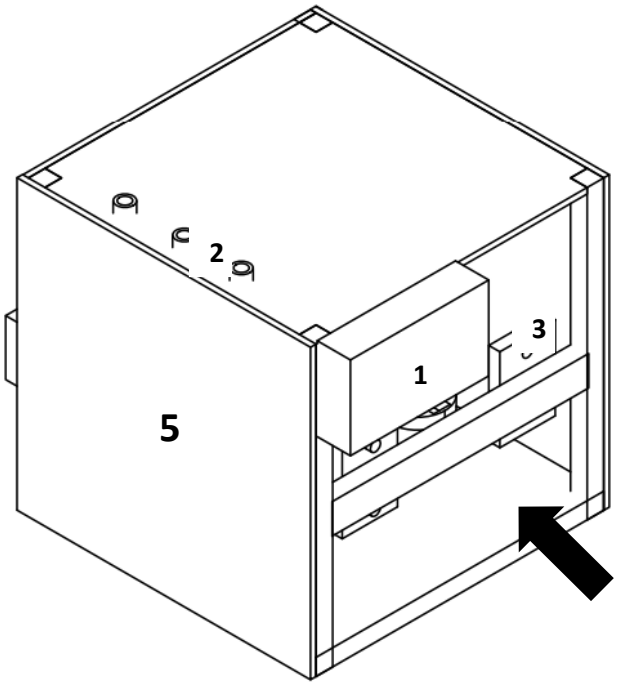
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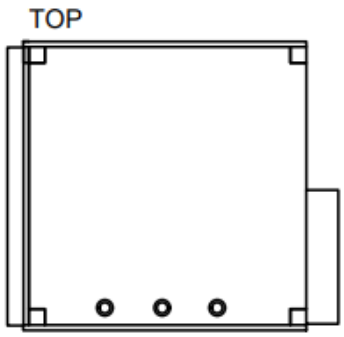
1. Introduction

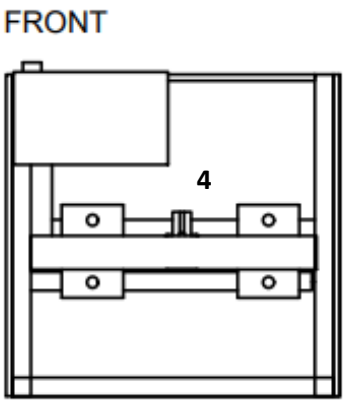
This manual provides detailed instructions for the safe and effective operation of the Automated Shoe Polishing Machine. The machine is designed to efficiently clean and polish formal leather shoes (loafers).

1.1 System Overview

The system is controlled by an Arduino Uno Microcontroller, which manages the DC motors for brush rotation, the NEMA 17 Stepper Motor for precise shoe rotation and brush translation, and a mini diaphragm pump for liquid polish dispensing.







PART LIST	
Number	**Name**
1	Control Box
2	Polish Tanks
3	Brush Systems
4	Shoe Holder System
5	Base

2. Safety and General Information

Category	Safety Precaution

Power & Electrical	The machine operates using a **12V DC adapter**. Do not expose the machine to water or attempt to operate it with wet hands. Always unplug the unit before performing maintenance.
Shoe Compatibility	This machine is specifically designed for **formal leather cover shoes**. Do not insert boots, high heels, or casual shoes, as they may cause mechanical jamming or damage the brushes.
Mechanical Safety	Keep fingers, hair, and clothing away from the rotating brushes and the shoe rotating mechanism during operation. Ensure the brush path is clear before starting the cycle.

3. Machine Overview and Status Indicators

The system communicates its status through an **LCD Display** and **LED Indicators** located on the front panel.

LED Indicator	Color	LCD Display	System State	Audio Feedb
Power	Red	-	Power inserted into the machine	-
System Active	Yellow	Shoe Polishing Machine	System is powered on, initialized, and awaiting color selection.	-
-	-	Select Color	Waiting for user input	-
-	-	Cleaning Shoe	Cleaning cycle active	-
-	-	Polishing Shoe	Polishing cycle active	-
-	-	Polishing Complete	The polishing cycle is finished and the shoe is ready for removal.	Alarm

4. Step-by-Step Operation Guide

Follow this sequence to operate the Automated Shoe Polishing Machine, as defined by the system flowchart.

Step	Action	User Input	Machine Response & Display
1. Power On	Plug the 12Vold DC adapter into the power socket.	None	**LCD:** Displays welcome message: "Shoe Polishing Machine". **LED:** Green LED turns ON (Ready).
2. Shoe Placement	Place your leather shoe (heel first) onto the shoe holder mechanism.	None	Ensure the shoe is securely positioned for the NEMA 17 motor to rotate it.
3. Color Selection	Press the button corresponding to your desired polish color.	Press **BLACK** or **BROWN** button[13].	**LCD:** Displays the selected color (e.g., "Color: Black Selected").
4. Start Cycle	Confirm the shoe is properly secured and the correct color is selected.	Press the **START** button.	**LED:** Blue LED turns ON (Polishing). The polishing sequence begins.
5. Polishing Cycle	Keep hands clear of the machine.	None	**Brushes:** DC Motors activate to rotate brushes. **Dispenser:** Pump activates to dispense liquid polish. The cycle runs for a predetermined period.
6. Cycle Complete	Wait for the audio notification.	None	**Motors/Pump:** Deactivate. **Buzzer:** Alarms completion. **LCD:** Remove Shoe.
7. Reset	Remove the polished shoe from the holder.	None	The system automatically resets to the initial state, ready for the next shoe.

5. Maintenance Routine

Category	Safety Precaution
Polish Refill	Regularly check the liquid polish reservoir for both colors. Use only high-quality, non-clogging liquid polish.
Nozzle Cleaning	Periodically clean the dispensing nozzle openings to prevent blockage.
Input voltage	